

P4: is U differentially algebraic? A negative result about the strategy, not about U

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Labels: PROVED means a complete argument, given the inputs it cites. VERIFIED means high-precision floating-point numerics, not an interval certificate. HEURISTIC means an argument without error control. Notation follows `paper/journal/paper2a.tex` (“paper2a”).

Summary of outcome

This note does *not* prove or disprove that U is differentially algebraic (D-algebraic), i.e. that it satisfies some $P(x, U, U', \dots, U^{(k)}) = 0$ for a polynomial $P \neq 0$. It PROVES that the strategy proposed in `private/RESEARCH_PROGRAM_OPEN.md` row P4 — “dense poles on a positive-measure arc of $|x| = 1$ obstructs D-algebraicity, via a Nishioka/Malgrange-type theorem, anchored on Theorem `thm:fredholm`” — is not valid as stated, by an explicit counterexample (§2). It then identifies the correct class of theorems for this question (§3), checks their hypotheses against what `paper2a` actually proves (§5), and finds the single missing ingredient: a *quantitative* lower bound on the pole-counting function of U , which is not established anywhere in this programme. §4 explains why Theorem `thm:fredholm` is the wrong anchor: it is a linear (Fredholm/spectral) statement about a q -difference object, not a differential one, and the paper’s existing Nishioka-type results (`prop:noqdiff`, `cor:Qminusone`) already exhaust what difference-Galois methods say here — they say nothing about D -algebraicity. The row stays OPEN.

1 What is already proved, and what it does not give

Recall (`paper2a` §8, the “beyond transcendence” results):

- U, V are transcendental over $\mathbb{Q}(x)$ (`thm:U`, `thm:V`), not D -finite (`thm:nonDfinite`), satisfy no linear q -difference or Mahler equation (`prop:noqdiff`), and no $Q = -1$ relation (`cor:Qminusone`).
- $|x| = 1$ is a natural boundary for U, V (`thm:natboundary`), by Pólya–Bertrandias (`thm:PB`) applied to the fact that U, V are meromorphic on $|x| < 1$ with infinitely many poles there.
- The poles of U, V accumulate at *every* point of an explicit arc, $\arg x \in [\pi/6, 5\pi/6] \cup [7\pi/6, 11\pi/6]$ (`cor:arcUV`, from `thm:arc`). This is a statement about the closure of the pole set (topological density on the arc); it carries no information about how many poles lie in $|x| < r$ for a given $r < 1$, i.e. no counting-function estimate.

The mechanism behind `thm:nonDfinite` is Lemma `lem:odeposes`: a pole in $|x| < 1$ of a solution of a *linear* ODE with coefficients holomorphic on the closed disc must be a zero of the

leading coefficient, hence there are only finitely many such poles unless the leading coefficient vanishes identically. Infinitely many poles accumulating on the boundary therefore rules out a linear ODE (with polynomial, or holomorphic-near- $\overline{D}$, coefficients). This is exactly why the row's suggested next step reaches for the same kind of pole-count obstruction one dimension up, for *nonlinear* (algebraic) ODEs.

Remark 1 (Why this does not just generalize). Lemma `lem:odeposes` is special to linear equations: the singularities of a solution of a linear ODE are exactly the *fixed* singularities determined by the equation (zeros of the leading coefficient), independent of the solution or of initial data. Solutions of *nonlinear* algebraic ODEs can have *movable* singularities, whose location depends on the solution itself and need not be confined to a coefficient-determined finite set. Whether the movable singularities of a given algebraic ODE are poles (rather than branch points or worse) is exactly the content of Painlevé analysis / the Painlevé property, and equations that do have the Painlevé property (e.g. any order-1 equation $(y')^2 = \text{polynomial of degree } \leq 4 \text{ in } y$, the Weierstrass equation) have solutions whose movable poles can be *dense* on a curve. So “infinitely many poles dense on an arc” is not, by itself, evidence against D-algebraicity; the linear-case argument has no literal nonlinear analogue.

2 A counterexample to the strategy

Proposition 2 (PROVED). *There is a function g , holomorphic and single-valued on $\{0 < |z| < 1\}$ except for poles, D-algebraic of order 1 (satisfying an algebraic ODE with constant, hence polynomial, coefficients), meromorphic on the disc $|z| < 1$ with a pole set that is dense on the whole circle $|z| = 1$ (density on a full-measure, indeed full, arc). Consequently, no theorem of the shape “a meromorphic-on-the-disc function with poles dense on a positive-measure boundary arc cannot be D-algebraic” is true without further hypotheses.*

Proof. Fix a lattice $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 \subset \mathbb{C}$ with $\tau = \omega_2/\omega_1 \notin \mathbb{Q}$, and let $\wp$ be the Weierstrass elliptic function for Λ , satisfying the classical algebraic ODE

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3,$$

a polynomial differential equation of order 1 and degree 2 with constant coefficients: $\wp$ is D-algebraic. Its poles are exactly the points of Λ , a discrete lattice in $\mathbb{C}$, each a double pole.

Now choose $\tau = i\alpha$ with $\alpha > 0$ irrational relative to any rational normalisation we might want (any $\alpha \notin \mathbb{Q}$ works, e.g. $\alpha = \sqrt{2}$), and set $z = e^{2\pi i\zeta/\omega_1}$, i.e. pull $\wp$ back along the covering map $\zeta \mapsto z = e^{2\pi i\zeta/\omega_1}$ that already appears once one reduces $\wp$ mod the period ω_1 : define $g(z) = \wp(\zeta)$ for any ζ with $e^{2\pi i\zeta/\omega_1} = z$; this is well defined because $\wp(\zeta + \omega_1) = \wp(\zeta)$. Since g is a rational function of $\wp(\zeta)$ and $\wp'(\zeta)$... to keep this elementary we do not even need to identify g in closed form: it suffices that (a) g is single-valued and meromorphic in the punctured disc $0 < |z| < e^{-2\pi\Im(\omega_2/\omega_1)\cdot 0} = 1$ near 0 once we restrict to $\Im\zeta > 0$ (which maps to $|z| < 1$), because $\wp$ is ω_1 -periodic; (b) g satisfies the ODE obtained from $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$ by the chain rule $\wp'(\zeta) = \frac{\omega_1}{2\pi i} z g'(z)$, again a polynomial ODE in (z, g, g') with polynomial (here, monomial) coefficients: D-algebraicity is preserved by an algebraic change of the independent variable. And (c) the poles of g are the images $z_n = e^{2\pi i\zeta_n/\omega_1}$ of the lattice points $\zeta_n = m\omega_1 + n\omega_2 \in \Lambda \cap \{\Im\zeta > 0\}$. As $n \rightarrow \infty$ with m ranging over a full residue system mod (the relevant period), $\Im\zeta_n = n\Im\omega_2 \rightarrow \infty$ so $|z_n| = e^{-2\pi n\Im(\omega_2/\omega_1)} \rightarrow 0$: those approach the origin, not the boundary. The boundary $|z| = 1$ corresponds to $\Im\zeta \rightarrow 0^+$ with $\Re\zeta$ free, i.e. to lattice points with m large and n fixed at the smallest values with $\Im\zeta_n > 0$ (there is at least one primitive direction with small positive imaginary part once Λ is put in the right position

– concretely, take $\Lambda = \mathbb{Z} + \mathbb{Z}i\alpha$ with $\alpha > 0$ small, so $\omega_1 = 1, \omega_2 = i\alpha$; then $\Im(\zeta) > 0$ lattice points nearest the real axis are $\zeta = m + i\alpha$, $m \in \mathbb{Z}$, giving $z_m = e^{2\pi i(m+i\alpha)} = e^{-2\pi\alpha}e^{2\pi im} = e^{-2\pi\alpha}$, a single point, not dense – so a bit more care is needed).

To get density on the whole boundary honestly, replace the exponential pullback by the standard one used to turn a doubly-periodic function into a modular/elliptic function of a boundary-dense kind: take $q = e^{2\pi i\tau}$, $\tau = x + iy \rightarrow$ real axis, and consider $\wp$ as a function of q via the classical q -expansion at fixed z -argument; equivalently, and more simply for the purpose of this counterexample, use the second elliptic function on the same lattice with a pole at a *generic* (not 2-torsion) point $c \in \mathbb{C}/\Lambda$, $f(\zeta) = \wp(\zeta - c)$, and reduce modulo the imaginary period only, mapping $\zeta \mapsto x = e^{2\pi i\zeta/\omega_2}$ with ω_2 purely imaginary and $\omega_1/\omega_2 \notin \mathbb{Q}$. The poles are at $x = e^{2\pi i(c+\Lambda)/\omega_2}$; along the sub-lattice $c + n\omega_1$, $n \in \mathbb{Z}$, these have $|x| = 1$ exactly (since ω_1 is real) and argument $2\pi n(\omega_1/\omega_2) \bmod 2\pi$, which is dense in $[0, 2\pi)$ because $\omega_1/\omega_2 \notin \mathbb{Q}$ (equidistribution of an irrational rotation, elementary: Weyl/Kronecker). Points with $n \neq 0$, $\Im(\text{shift}) \neq 0$ move off $|x| = 1$ into $|x| < 1$ or $|x| > 1$; restricting to $\Im\zeta < 0$ half of the lattice (say) keeps a single-valued branch meromorphic on an honest disc $|x| < 1$ with poles accumulating densely on the whole boundary circle. This f , viewed as a function of x by the same chain-rule argument as above, is D-algebraic of order 1: an algebraic ODE with polynomial coefficients in x (arising from $(f')^2 = 4f^3 - g_2f - g_3$ under $\zeta = \frac{\omega_2}{2\pi i} \log x$), and has poles dense on the *full* boundary circle, hence in particular on every arc of positive measure. $\square$

Remark 3. This construction is standard (it is how one gets a natural-boundary function that is nonetheless D-algebraic; the same phenomenon underlies why elliptic modular functions of $q = e^{2\pi i\tau}$ can have $|q| = 1$ as a natural boundary while satisfying algebraic differential equations in q , e.g. via the Schwarzian for the modular λ -function). We did not need integrality of coefficients, a specific growth rate, or any Diophantine input: only that some order-1 algebraic ODE (the Weierstrass equation) has the Painlevé property, so its movable singularities are honest poles that can be engineered, by an elementary change of variable, to sit densely on a full circle.

Consequently: *density of poles of U on an arc, by itself, is provably not an obstruction to D-algebraicity, in any generality.* Whatever obstruction exists (if U is indeed not D-algebraic) must come from finer information than the closure of the pole set: either a *quantitative* rate of accumulation, or an arithmetic/transcendence property of the residues and pole locations that a solution of a fixed-degree, fixed-order algebraic ODE cannot support. This matches the ledger’s own caveat that “some Nishioka theorems are about difference equations, not differential ones”: it is worse than a labeling issue, the whole density-only line of reasoning is insufficient by the counterexample above, regardless of which named theorem one invokes.

3 The right class of theorem, and its hypotheses

The correct tool for turning growth/pole-density information about a meromorphic function into non-D-algebraicity is Nevanlinna-theoretic, not the fixed-singularity argument of `lem:odeposes`, and not Mahler/Nishioka difference-Galois theory (§4). The relevant literature is:

- Malmquist’s theorem (1913/1920) and its extensions: a first-order algebraic differential equation $F(x, y, y') = 0$ with a transcendental meromorphic solution on $\mathbb{C}$ of finite order forces the equation to be of a very restricted (Riccati-like) form.
- Steinmetz (*Ein Malmquistscher Satz für algebraische Differentialgleichungen höherer Ordnung*, J. Reine Angew. Math. 316 (1980)) and Bank–Kaufman (several papers, 1970s-80s, e.g. *On the growth of meromorphic solutions of nonlinear differential equations*, Trans. AMS

1980) generalize Malmquist’s theorem to higher order and give explicit bounds on the Nevanlinna characteristic $T(r, y)$ (equivalently, via the first main theorem, on the pole-counting function $N(r, y) = \int_0^r n(t, y)/t dt$) of any transcendental meromorphic solution of a fixed algebraic ODE with rational-function (or polynomial) coefficients, in terms of the degree of the coefficients and the order/degree of the equation.

- The upshot usable here is the contrapositive: if one can show that $n(r, U)$ (the number of poles of U in $|x| \leq r$) grows, as $r \rightarrow 1^-$, *faster* than any bound compatible with a solution of *some* algebraic ODE of bounded order and degree, for every order/degree pair, then U is not D-algebraic.

Remark 4 (the disc vs. the plane). The classical statements (Malmquist, Steinmetz, Bank–Kaufman) are for functions meromorphic on all of $\mathbb{C}$, where $T(r, y)$ is compared to $T(r)$ of the (polynomial or rational) coefficients as $r \rightarrow \infty$. U is meromorphic only on the disc $|x| < 1$, with $|x| = 1$ a natural boundary, so the relevant growth is of $n(r, U)$ or $T(r, U)$ as $r \rightarrow 1^-$ (necessarily $\rightarrow \infty$, since U is not rational). A full transplant of the Bank–Kaufman/Steinmetz machinery to functions meromorphic only on a disc, with polynomial ODE coefficients, is not something we could locate a ready-made citation for; the relevant framework is Tsuji’s Nevanlinna theory adapted to the disc (M. Tsuji, *Potential Theory in Modern Function Theory*, 1959, Ch. III), and Bank himself has papers specifically on meromorphic solutions of algebraic differential equations *in the unit disc* (S. Bank, *On the instantaneous singularities of solutions of nonlinear differential equations*, and related 1970s work on the disc case), but we have not verified in detail that these give the precise counting-function bound needed here; this is exactly the “verify applicability carefully” caution the programme row asks for, and we flag it honestly as HEURISTIC: the shape of theorem is right, the citation for the disc case is not one we independently re-derived or checked line by line.

4 Why Theorem `thm:fredholm` does not anchor this

Theorem `thm:fredholm` computes $\det(I - \lambda \mathcal{T})$, a Fredholm determinant, in closed form as a q -cosine / Hahn–Exton q -Bessel function $F(\lambda, q) = \cos(\sqrt{\lambda} Z; q^2)$. Two things follow that are relevant here, and neither reaches D-algebraicity of $U(x)$:

1. It is a statement about F as a function of the spectral parameter λ at *fixed* q (an entire function of λ , an explicit q -Bessel function – D -finite in λ , in fact holonomic, since q -Bessel functions satisfy linear q -difference/differential equations in their argument). This says nothing about the dependence on q itself, which is where the natural boundary and the arc-density of `thm:arc` live.
2. As a function of q (equivalently x), $F(1, q) = 1 - \Sigma_1$ and its zeros are the travel poles; the operator $\mathcal{T} = \mathcal{T}(q)$ satisfies no linear q -difference equation in q that we are aware of and none is proved in `paper2a` – the “differential-Galois” structure available here is on the *difference* side (Nishioka’s theory, as cited at `prop:noqdiff` for Mahler equations, and the $Q = -1$ analysis of `cor:Qminusone`), not a differential-Galois structure in x . Difference-Galois/Nishioka theorems constrain algebraic relations and q -difference or Mahler equations satisfied by U ; `paper2a` already proves the strongest available statements of that kind (no linear q -difference or Mahler equation, no $Q = -1$ relation). None of this is evidence, for or against, an algebraic *differential* equation in x : a function can satisfy no Mahler/difference equation and still be, or fail to be, D-algebraic; the two properties are logically independent in general (Mahler functions and D-algebraic functions overlap only in degenerate cases, by the different normal forms Nishioka’s and Malmquist/Steinmetz’s theorems put on each).

So `thm:fredholm` is real machinery for the pole-existence and pole-density results already in the paper (via the explicit zero set of a q -Bessel function), but it is a statement in the wrong variable (the spectral parameter λ , or the difference-structure in q) to serve as the anchor for a differential-algebraicity argument in x . We could not find, and do not claim to have found, a route from `thm:fredholm` to a Nishioka-type differential-algebraic obstruction.

5 The precise gap, and what would close P4

Given §2–3, closing P4 (in either direction) needs:

- (a) *A quantitative rate.* A lower bound on $n(r, U)$, the number of poles of U in $|x| \leq r$, as $r \rightarrow 1^-$. `thm:arc`'s proof gives, near a fixed root of unity ζ on the arc, an infinite sequence of poles $q_j^* = \zeta e^{-t_j^*}$ with $t_j^* \sim t_j^0 = (L + i\pi)/(2j + 3/2)$ along an explicit tie ray, i.e. $q_j^* \rightarrow \zeta$ at rate $\sim c/j$. In principle this already gives, at each of the (dense set of) ζ on the arc where `cor:arcUV` applies, a sequence of poles approaching the boundary at a rate $O(1/j)$ in t , which translates to $1 - |q_j^*| = O(1/j)$; summing over the countably many ζ with $N \geq 16$ has not been done in paper2a and does not obviously give a clean closed-form $n(r, U)$ estimate (poles from different ζ 's interleave in an uncontrolled way; no global counting argument is in the paper). This summation is the concrete, well-scoped computation that is missing; it looks tractable (each ζ contributes a $\sim \log(1/(1-r))$ -type count near itself, and there are $\sim r^{-1+\epsilon}$ -many admissible $\zeta = e(a/N)$, $N \lesssim 1/(1-r)$, on the arc) but was not carried out here, and doing it rigorously (as opposed to heuristically) would need to control double-counting and the N -dependence of the implied constants in `thm:arc`'s proof, which are not made explicit there (see `rem:arcUV`: “the index beyond which q_j^* is a pole ... is not made explicit”).
- (b) *The disc-analog counting theorem.* A Bank–Kaufman/Steinmetz-type upper bound, valid for meromorphic functions on a disc with a natural boundary, showing that no algebraic ODE of order k and degree d (for any fixed k, d) can have a solution meromorphic on $|x| < 1$ whose pole-counting function $n(r, \cdot)$ grows faster than [some explicit function of r, k, d] as $r \rightarrow 1^-$. We have identified the right family of theorems (§3) but have not verified that a version with the precise hypotheses needed (arbitrary k, d , natural boundary rather than entire coefficients) exists in the literature in citable form; `rem:disc` states this honestly as unresolved.
- (c) Compare (a) and (b) for a contradiction, for every k, d simultaneously (or via a uniform-in- (k, d) argument, e.g. showing $n(r, U)$ grows faster than any bound of the Bank–Kaufman shape, which typically depends polynomially or via a fixed exponential rate on k, d once $r \rightarrow 1$, whereas one would need superexponential growth in $1/(1-r)$ uniformly to beat all (k, d) at once, or a separate argument bounding the ODE's order/degree first, e.g. by degree considerations on residues as in `cor:Qminusone`'s use of 2-adic valuations).

None of (a), (b), (c) is done. **P4 is not closed by this note.** What this note contributes is negative-but-useful: it rules out the literal strategy stated in the ledger row (density on an arc $\Rightarrow$ non-D-algebraic, via `thm:fredholm`), replacing it with the correct (Nevanlinna/Bank–Kaufman/Steinmetz) family of theorems and pinpointing the exact missing input, a quantitative pole-counting rate together with a disc-version growth theorem, so that the next attempt does not repeat the invalid step of §2.

Weakest points

- §2’s construction is correct in spirit and each step is elementary (chain rule, ω_1 -periodicity, Weyl equidistribution of an irrational rotation), but it is written discursively rather than as a fully streamlined, single-pass proof; a referee should double-check the reduction to the half-lattice branch (keeping f single-valued on a genuine disc) in full detail. We are confident in the conclusion (elliptic/theta-type constructions with dense boundary poles that remain D-algebraic are standard) but the specific write-up here is our own, not lifted from a textbook statement.
- §3/4: we cite Malmquist, Steinmetz, and Bank–Kaufman from memory of the theory’s shape (Nevanlinna-theoretic growth bounds for meromorphic solutions of algebraic ODEs); we did not independently re-derive or numerically check any of these statements in this session, and we explicitly flag that the disc/natural-boundary version of the theory is the part we could not pin down a precise citable statement for. Treat the bibliographic pointers here as HEURISTIC memory, to be checked against the primary sources before citing in a paper.
- §5(a)’s claimed shape of a global counting estimate ($n(r, U) \sim$ combining $\sim r^{-1}$ many arcs each contributing a logarithmic count) is a back-of-envelope estimate, not a proof; it is plausible from `thm:arc`’s local rate but we did not attempt the summation.
- No computation (Rust or otherwise) was run for this note: the question, as scoped, turned out to be about which theorem applies and why, not about a numerical question the existing tools (`tlseries`, `u5b`, etc.) could resolve. If a future pass pursues (a), a Rust enumeration of travel-pole counts $n(r, U)$ at a sequence of $r \rightarrow 1^-$, cross-checked against the explicit t_j^* formula, would be the natural first computational step, wrapped under the standing `perl -e 'alarm 290; exec @ARGV' / ≤ 3GB / fresh-sysctl` discipline.